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stable-diffusion-xl-lightning

Text-to-Image • bytedance

Beta

@cf/bytedance/stable-diffusion-xl-lightning

SDXL-Lightning is a lightning-fast text-to-image generation model. It can generate high-quality 1024px images in a few steps.

Model Info

More information	link ↗
Beta	Yes
Unit Pricing	\$0.00 per step

Usage

✓ Workers - TypeScript

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const inputs = {  
      prompt: "cyberpunk cat",  
    };  
  
    const response = await env.AI.run(  
      "@cf/bytedance/stable-diffusion-xl-  
lightning",  
      inputs  
    );  
  
    return new Response(response, {  
      headers: {  
        "content-type": "image/jpg",  
      },  
    });  
  },  
}
```

```
} satisfies ExportedHandler<Env>;
```

▼ curl

```
curl
https://api.cloudflare.com/client/v4/accounts/
diffusion-xl-lightning \
-X POST \
-H "Authorization: Bearer $CLOUDFLARE_API_TOKEN" \
-d '{ "prompt": "cyberpunk cat" }'
```

Parameters

* indicates a required field

Input

- prompt **string** required min 1

A text description of the image you want to generate

- negative_prompt **string**

Text describing elements to avoid in the generated image

- **height** **integer** min 256 max 2048

The height of the generated image in pixels

- **width** **integer** min 256 max 2048

The width of the generated image in pixels

- **image** **array**

For use with img2img tasks. An array of integers that represent the image data constrained to 8-bit unsigned integer values

- **items** **number**

A value between 0 and 255

- **image_b64** **string**

For use with img2img tasks. A base64-encoded string of the input image

- **mask** **array**

An array representing An array of integers that represent mask image data for inpainting constrained

to 8-bit unsigned integer values

- `items` **number**

A value between 0 and 255

- `num_steps` **integer** default 20 max 20

The number of diffusion steps; higher values can improve quality but take longer

- `strength` **number** default 1

A value between 0 and 1 indicating how strongly to apply the transformation during img2img tasks; lower values make the output closer to the input image

- `guidance` **number** default 7.5

Controls how closely the generated image should adhere to the prompt; higher values make the image more aligned with the prompt

- `seed` **integer**

Random seed for reproducibility of the image generation

Output

The binding returns a `ReadableStream` with the image in JPEG or PNG format (check the model's output schema).

API Schemas

The following schemas are based on JSON Schema

Input

Output

```
{  
  "type": "object",  
  "properties": {  
    "prompt": {  
      "type": "string",  
      "minLength": 1,  
      "description": "A text description  
of the image you want to generate"  
    },  
    "negative_prompt": {  
      "type": "string",  
      "description": "Text describing  
elements to avoid in the generated image"  
    },  
    "height": {  
      "type": "integer",  
      "minimum": 256,  
      "maximum": 2048,  
      "description": "The height of the  
generated image in pixels"  
    },  
    "width": {  
      "type": "integer",  
      "minimum": 256,
```

```
    "maximum": 2048,  
    "description": "The width of the  
generated image in pixels"  
  },  
  "image": {  
    "type": "array",  
    "description": "For use with  
img2img tasks. An array of integers that  
represent the image data constrained to 8-bit  
unsigned integer values",  
    "items": {  
      "type": "number",  
      "description": "A value  
between 0 and 255"  
    }  
  },  
  "image_b64": {  
    "type": "string",  
    "description": "For use with  
img2img tasks. A base64-encoded string of the  
input image"  
  },  
  "mask": {  
    "type": "array",
```

```
    "description": "An array  
representing An array of integers that  
represent mask image data for inpainting  
constrained to 8-bit unsigned integer values",  
    "items": {  
        "type": "number",  
        "description": "A value  
between 0 and 255"  
    }  
},  
"num_steps": {  
    "type": "integer",  
    "default": 20,  
    "maximum": 20,  
    "description": "The number of  
diffusion steps; higher values can improve  
quality but take longer"  
},  
"strength": {  
    "type": "number",  
    "default": 1,  
    "description": "A value between 0  
and 1 indicating how strongly to apply the  
transformation during img2img tasks; lower
```

values make the output closer to the input image"

```
    },  
    "guidance": {  
      "type": "number",  
      "default": 7.5,  
      "description": "Controls how  
closely the generated image should adhere to  
the prompt; higher values make the image more  
aligned with the prompt"  
    },  
    "seed": {  
      "type": "integer",  
      "description": "Random seed for  
reproducibility of the image generation"  
    }  
  },  
  "required": [  
    "prompt"  
  ]  
}
```

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